FSC repeatability test

Not repeatable Without the fix, **500** runs on the same website text of Chicago Nut & Bolt returned **11** different sets of FSC codes, and only **56.6**% matched the most common set. Repeatable With the fix, **200** runs returned **1** set, in **1** order.

Before and after the fix

Each setup ran on the same frozen website text. The seed alone made Gemini repeatable; the caches make every repeat exact, including Jev's small run-to-run noise that swaps two near-equal top codes.

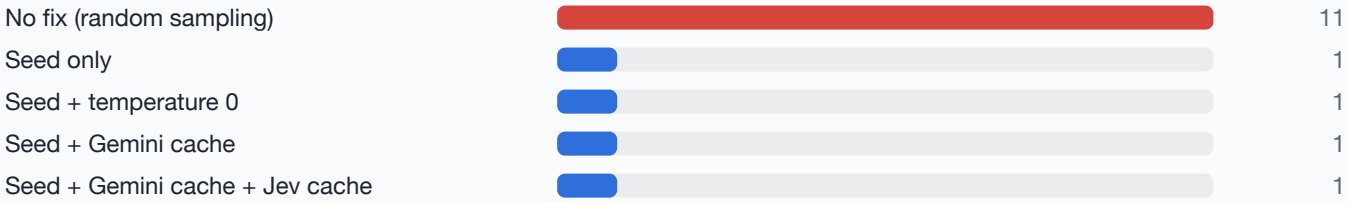
Setups compared

The first row is the run without the fix; each later row adds one layer.

Setup	Runs	Code sets	Same as most common %	Orderings	Gemini profiles	Median run s
No fix (random sampling)	**500**	**11**	**56.6**	**61**	**185**	**4.3**
Seed only	100	1	100.0	2	1	4.6
Seed + temperature 0	100	1	100.0	2	1	4.7
Seed + Gemini cache	100	1	100.0	2	1	1.0
Seed + Gemini cache + Jev cache	200	1	100.0	1	1	0.5

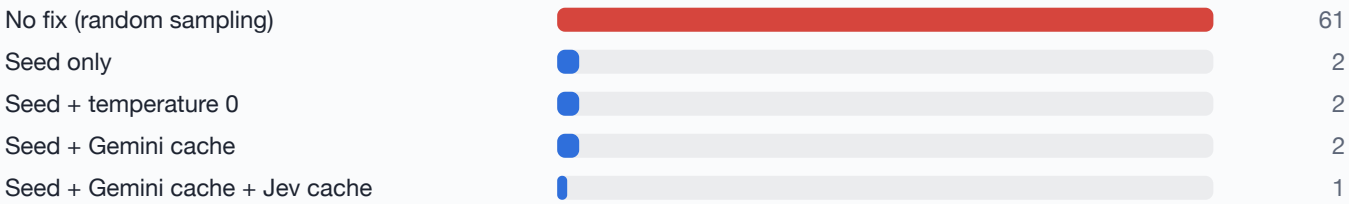
Different code sets

Fewer is better. One means every run returned the same codes.



Different orderings of the list

The same codes can still rank differently. One means the list is identical every time.



Without the fix, in detail

Runs completed
500
Same company, same website text

Different code sets
11
Expected: identical every time

Match the most common set
56.6 %
Share of all runs

Different orderings
61
Same codes can still rank differently

Median run time
4.3 s
Full pipeline, live Gemini and Jev

Is the answer the same every time?

Code sets returned

Every distinct set of codes, most common first. Codes in red are not in the most common set.

#	Codes	Runs	Share %	
1	**5305, 5306, 5310, 5320, 5325, 5340, 5365**	**283**	**56.6**	
2	5305, 5306, **5307**, 5310, 5320, 5325, 5340, 5365	155	31	
3	5305, 5306, 5310, 5320, 5325, 5340	28	5.6	
4	5305, 5306, 5310, 5320, 5325, 5340, **5342**, 5365	10	2	
5	5305, 5306, **5307**, 5310, 5320, 5325, 5340	9	1.8	
6	5305, 5306, **5307**, 5310, 5320, 5325, 5340, 5365, **9510**	5	1	
7	5305, 5306, **5307**, 5310, 5320, 5325, 5340, **5342**, 5365	4	0.8	
8	5305, 5306, **5307**, 5310, 5320, 5325, 5340, **5342**	3	0.6	
9	5305, 5306, **5307**, 5310, 5320, 5325, 5340, **5342**, 5365, **9510**	1	0.2	
10	5305, 5306, 5310, 5325, 5340, **5342**	1	0.2	
11	5305, 5306, **5307**, 5310, 5320, 5325, 5340, **9510**	1	0.2	

How often each code appears

Share of runs that returned the code. Red codes come and go between runs.

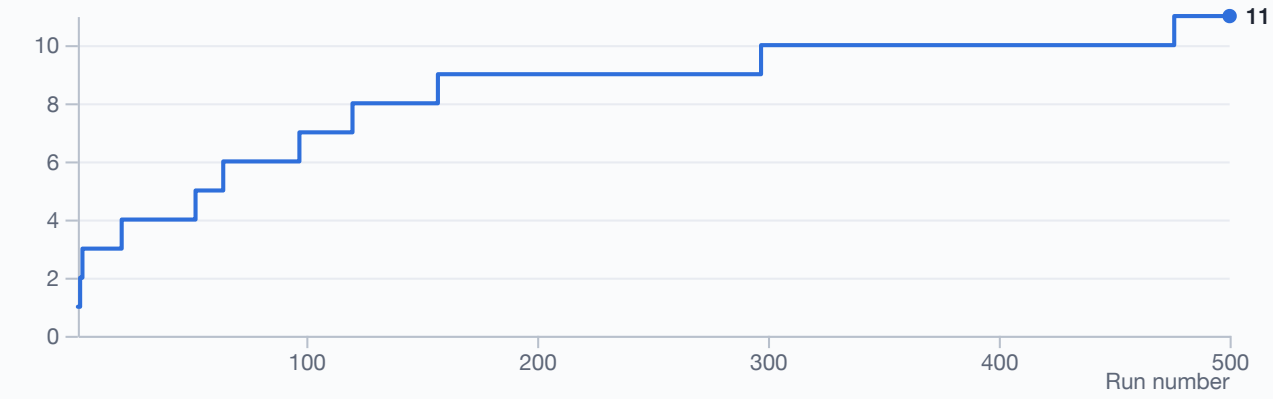
In nearly every run Comes and goes

5306 Bolts	100.0%
5310 Nuts and Washers	100.0%
5305 Screws	100.0%
5325 Fastening Devices	100.0%
5340 Hardware, Commercial	100.0%
5320 Rivets	99.8%
5365 Bushings, Rings, Shims and Spacers	91.6%
5307 Studs	35.6%
5342 Hardware, Weapon System	3.8%
9510 Bars and Rods	1.4%

How quickly the differences show up

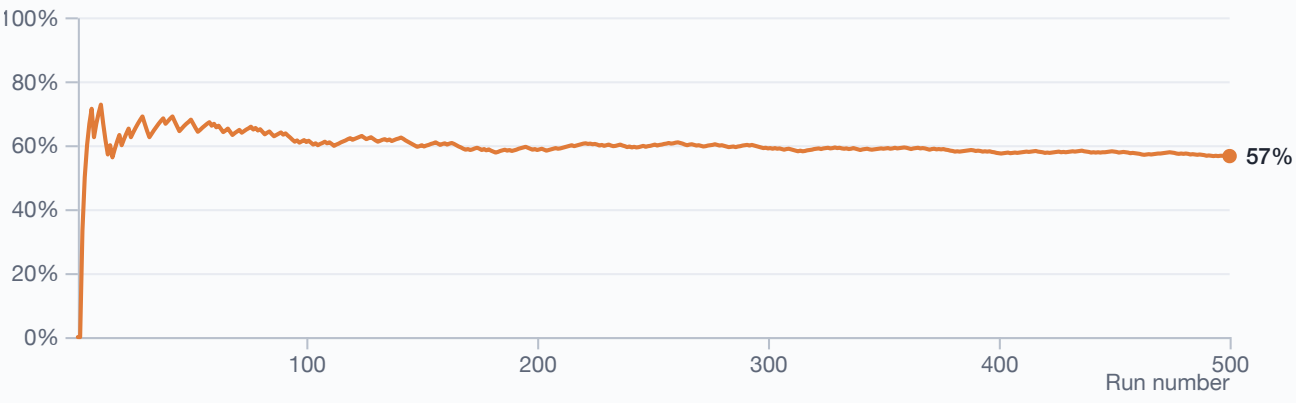
Different code sets seen so far

Counts new sets as the runs accumulate. A repeatable pipeline would stay flat at one.



Runs matching the most common set

Running share, in percent. It settles, but well below the whole.



Where the differences start

Of the **217** runs that differed from the most common answer, **82**% started with a different profile written by Gemini and **17.5**% with a different list of codes proposed by Gemini.

First stage where a run departs from the most common run

Each differing run is counted once, at the earliest stage whose output differs. Jev's own scoring is not isolated here, because it reads the whole profile and that text changes from run to run.



Gemini's product-line profile	**82.0% · 178 runs**
Gemini's proposed code list	**17.5% · 38 runs**
Jev scoring and ranking	**0.5% · 1 runs**

What Gemini writes from identical text

The website text was crawled once and reused. Across **500** runs Gemini wrote **185** different product-line profiles from it; the most common one appeared in **20.2**% of runs.

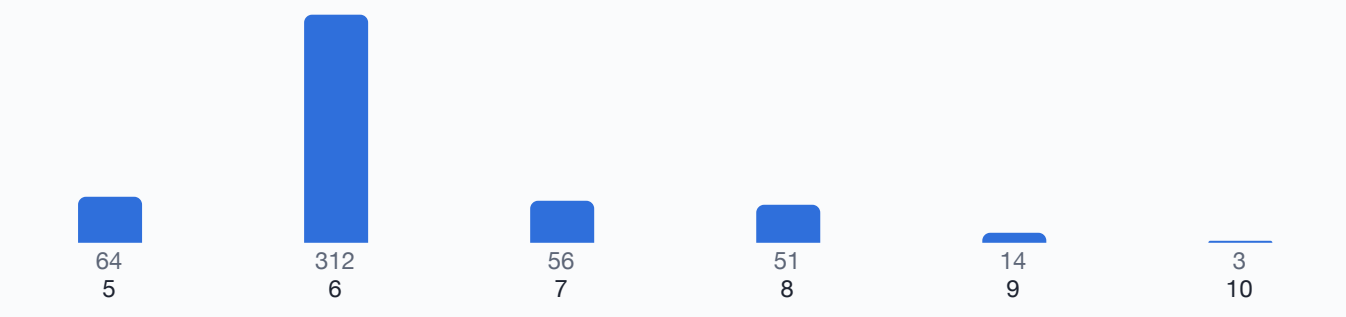
Product-line names it used

Share of runs that contained each exact name.



Product lines per profile

Number of runs by how many product lines Gemini listed.



How the run without the fix was set up

Company Chicago Nut & Bolt	Website https://cnb-inc.com/	Pages crawled once 6
Characters of text 29,275	Gemini model gemini-3.5-flash-lite	Decider jev-1.13.0
Sampling seed none	Temperature model default	Output cache off
Parallel runs 6		

Produced by `backend/scripts/determinism.py`. The site is crawled once and its text frozen, so differences come from the models and our own code, not from the site changing. Every run records its profile, candidate pool and Jev probabilities so the first stage that diverges can be named.

Static export of the interactive dashboard "FSC repeatability test". Figures come from backend/scripts/determinism.py runs on Chicago Nut & Bolt (cnb-inc.com); the website text was crawled once and frozen for every run.